BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

# How hormones work

The written notes for the how hormones work walkthrough. Week 8 competencies 7 to 11.

**To print:** these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

**How to use these notes.** Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

## Step by step: from gland to target cell

**In your textbook:** Silverthorn Chapter 8, the parts on the classification of hormones and on control of hormone release.

### A peptide hormone, step by step (insulin as the example)

1. Ribosomes on the rough ER build a long, inactive chain, the preprohormone.
2. Inside the ER it is trimmed to a prohormone.
3. The Golgi apparatus packages the prohormone into secretory vesicles, where enzymes cut it into the active hormone. The vesicles are stored in the cell.
4. A trigger arrives. For insulin it is a rise in blood glucose, which the beta cells of the pancreas sense directly.
5. The vesicles fuse with the cell membrane and release the hormone by exocytosis.
6. The hormone dissolves in plasma and travels free. Insulin's half life is about 5 minutes.
7. Free hormone leaves the capillary and binds a receptor on the surface of the target cell.
8. The receptor acts through a second messenger or an enzyme, changing proteins the cell already has, within seconds to minutes.

### A steroid hormone, step by step (cortisol as the example)

1. A trigger arrives at the cell.
2. Enzymes in the smooth ER and mitochondria convert cholesterol into the hormone, one step at a time. Nothing is stored.
3. The hormone diffuses out through the cell membrane as soon as it is made.
4. In the blood, most molecules bind carrier proteins. For cortisol, only about 5 to 10 percent is free.
5. Only the free hormone leaves the capillary. As it does, bound molecules let go of their carriers to replace it.
6. Bound hormone is protected from breakdown and filtering, so the half life is longer: about an hour for cortisol.
7. The free hormone diffuses into the target cell and binds a receptor in the cytoplasm or nucleus.
8. The hormone and receptor turn a gene on or off. The cell makes messenger RNA and a new protein, so the response takes 90 minutes or more.

### Three triggers for release, step by step

1. Humoral: blood Ca2+ falls, parathyroid cells sense it and release PTH, PTH acts on bone and kidney, blood Ca2+ rises, and PTH release slows (negative feedback).
2. Neural: a sympathetic neuron from the spinal cord releases acetylcholine onto chromaffin cells in the adrenal medulla, which release epinephrine (and some norepinephrine) into the blood within seconds.
3. Hormonal: TSH from the anterior pituitary travels in the blood to the thyroid, which releases thyroid hormone (T3 and T4). The next walkthrough covers where TSH comes from.

## Three kinds of hormone

**In your textbook:** Silverthorn Chapter 8, the parts on hormones and on the classification of hormones.

**Hormone**

A chemical made by specialized cells, released into the blood, and carried to distant target cells, where it acts at very low concentrations.

**Endocrine cell**

A cell that makes and releases a hormone. Many sit together in endocrine glands, such as the thyroid, adrenal glands and pancreas.

**Target cell**

A cell with a receptor for the hormone. The blood carries a hormone past almost every cell, but only target cells respond.

| | Peptide | Steroid | Amine |
|---|---|---|---|
| **Made from** | Amino acids, as a preprohormone, then a prohormone | Cholesterol | One amino acid: tyrosine (catecholamines, thyroid hormones) or tryptophan (melatonin) |
| **Where it is made** | Rough ER, packaged by the Golgi | Smooth ER and mitochondria | Cytoplasm (catecholamines); thyroid follicle (thyroid hormones) |
| **Stored?** | Yes, in secretory vesicles, made ahead | No, made when needed | Catecholamines: yes, in vesicles. Thyroid hormones: stored in the thyroid gland, not in vesicles |
| **How it leaves the cell** | Exocytosis | Diffuses through the membrane | Catecholamines: exocytosis. Thyroid hormones: diffuse out |
| **Dissolves in** | Water (lipophobic) | Lipid (lipophilic) | Catecholamines: water. Thyroid hormones: lipid |
| **Receptor** | On the cell surface | In the cytoplasm or nucleus | Catecholamines: surface. Thyroid hormones: nucleus |
| **Response** | Seconds to minutes, by changing existing proteins | 90 minutes or more, by changing genes and making new protein | Catecholamines like peptides; thyroid hormones like steroids |
| **Examples** | Insulin, glucagon, parathyroid hormone | Cortisol, aldosterone, estrogen, progesterone, testosterone | Epinephrine, norepinephrine, dopamine, thyroid hormones, melatonin |

- Most hormones are peptides.
- Surface receptors for peptides mostly pass the message inside through second messengers such as cAMP, or act as enzymes, as you saw in Week 2.
- Some steroids also bind membrane receptors and cause fast effects that do not involve genes, but the classic steroid action is the slow one through genes.
- Amines do not behave as one class. What decides how each one behaves is the molecule itself: catecholamines dissolve in water, and thyroid hormones (two tyrosines with iodine attached) dissolve in lipid.

## Traveling in the blood

**In your textbook:** Silverthorn Chapter 8, the parts on the classification of hormones and on hormone transport and half life.

- Peptides and catecholamines dissolve in plasma and travel free.
- Steroids and thyroid hormones dissolve poorly in water, so most of their molecules travel bound to carrier proteins. Some carriers are general, such as albumin; some carry one hormone, such as thyroxine-binding globulin.
- Only free hormone can leave the capillary and bind a receptor. The carrier is too large to leave the blood.
- Bound hormone is a reserve. As free hormone leaves the blood, bound molecules let go of their carriers to replace it.
- Free fraction: about 5 to 10 percent of plasma cortisol; well under 1 percent of thyroxine.

**Half life**

The time it takes for half of a hormone to disappear from the blood.

**How hormones are cleared**

Enzymes in the blood and tissues break them down, the liver changes them into forms the body can get rid of, and the kidneys filter free hormone into the urine.

| Hormone | Carried | Approximate half life |
|---|---|---|
| **Insulin (peptide)** | Free | About 5 minutes |
| **Cortisol (steroid)** | Mostly bound | About an hour |
| **Thyroxine (thyroid hormone)** | Almost all bound | About a week |

- Binding protects a hormone from the enzymes that break it down, and makes it too large for the kidneys to filter, so bound hormones last longer.
- A short half life lets the body change a hormone's level quickly; a long half life gives a steady level that changes slowly.

### More carrier protein, step by step

1. The carrier protein level rises, as thyroxine-binding globulin does in pregnancy or with estrogen.
2. The new carriers bind free hormone, so free hormone falls.
3. Feedback control responds to free hormone, so the gland releases more.
4. Free hormone returns to normal; total hormone is now higher.
5. The person feels well. When carrier levels may have changed, the useful lab number is the free hormone level, not the total.

## Receptors decide the response

**In your textbook:** Silverthorn Chapter 8, the part on hormones and their receptors, with the review of receptors in Chapter 6.

The response to a hormone depends on which receptor the target cell has and which pathway that receptor turns on. The same hormone can cause opposite responses in different tissues.

| Tissue | Receptor | What epinephrine does |
|---|---|---|
| **Smooth muscle of the airways** | Beta-2 | Relaxes, so the airway widens |
| **Smooth muscle of small arteries in the skin and gut** | Alpha-1 | Contracts, so the artery narrows |
| **Heart muscle** | Beta-1 | Beats faster and more forcefully |
| **Small arteries in skeletal muscle** | Beta-2 | Relaxes, so the artery widens |

- Beta receptors raise cAMP. Alpha-1 receptors raise Ca2+ inside smooth muscle cells.
- Together these shift blood and air toward working muscles during exercise or danger.

**Down-regulation**

Long-lasting high hormone, over hours to days, makes the target cell remove receptors. The dose-response curve shifts right: the cell is less sensitive and needs more hormone for the

same response. Part of the reason some drugs lose their effect with heavy use (tolerance).

**Up-regulation**

Weeks of low hormone make the target cell add receptors. The curve shifts left: the cell is more sensitive.

**Dose-response curve**

A graph of how strongly a target cell responds at each hormone concentration.

## Hormones working together

**In your textbook:** Silverthorn Chapter 8, the part on hormone interactions.

**Synergism**

The effect of two or more hormones together is greater than the sum of their separate effects. Example: glucagon, epinephrine and cortisol on blood glucose. Each hormone acts through its own receptor and pathway.

**Permissiveness**

One hormone cannot exert its full effect unless a second hormone is present, even though the second does little on its own. Example: thyroid hormone and the reproductive hormones in the maturation of the reproductive system.

**Functional antagonism**

Two hormones have opposite effects through different receptors and pathways. Example: insulin and glucagon on the liver.

| Given | Maturation of the reproductive system |
|---|---|
| **Reproductive hormones, no thyroid hormone** | Delayed |
| **Thyroid hormone alone** | None |
| **Both** | Normal |

- Thyroid hormone also increases the number of beta receptors in tissues such as the heart, which lets epinephrine have its full effect; too much thyroid hormone can bring a racing heart.
- Synergy among glucagon, epinephrine and cortisol is part of why blood glucose climbs during severe stress or illness.
- Insulin binds a receptor enzyme and tells liver cells to take up glucose and store it as glycogen, so blood glucose falls. Glucagon binds a G protein coupled receptor and tells them to break glycogen down and release glucose, so blood glucose rises.
- Competitive antagonism, two molecules competing for the same receptor, is a different mechanism, used by many drugs.

## What tells a gland to release

**In your textbook:** Silverthorn Chapter 8, the part on control of hormone release.

OpenStax calls the three triggers humoral, neural and hormonal stimuli. Silverthorn describes the same three as the starting points of endocrine reflex pathways.

| Stimulus | What it is | Examples |
|---|---|---|
| **Humoral** | The endocrine cell itself senses a change in the blood | Low blood Ca2+ releases parathyroid hormone (PTH); high blood glucose releases insulin |
| **Neural** | A neuron signals the endocrine cell | Sympathetic neurons release acetylcholine onto chromaffin cells of the adrenal medulla, which release epinephrine; the posterior pituitary (next walkthrough) |
| **Hormonal** | Another hormone, a trophic hormone, tells the gland to release | TSH from the anterior pituitary releases thyroid hormone (T3 and T4) |

- Four small parathyroid glands sit on the back of the thyroid. PTH acts on bone to release Ca2+ into the blood and on the kidneys to keep more Ca2+ out of the urine.
- When the response removes the stimulus, as rising Ca2+ slows PTH release, that is negative feedback.
- The adrenal medulla, the inner part of the adrenal gland, is a modified part of the sympathetic nervous system. Its response comes within seconds.
- Trophic hormone: a hormone that tells another endocrine gland to release its hormone. The next walkthrough follows these chains from the hypothalamus through the pituitary.